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Zij $g(x, y) = e^x \sin y$ en zij $f(x, y) = (xy^2, x^2y)$

Eerst voor $g(x, y)$:

$$\frac{\partial g}{\partial x} = e^x \sin y$$

$$\frac{\partial g}{\partial y} = e^x \cos y$$

Daarna voor $f(x, y)$:

$$\frac{\partial f_1}{\partial x} = y^2$$

$$\frac{\partial f_2}{\partial x} = 2xy$$

$$\frac{\partial f_1}{\partial y} = 2xy$$

$$\frac{\partial f_2}{\partial y} = x^2$$

$$Dg(x, y) = \begin{pmatrix} e^x \sin y & e^x \cos y \end{pmatrix}$$

$$Df(x, y) = \begin{pmatrix} y^2 & 2xy \\ 2xy & x^2 \end{pmatrix}$$

Vul $f(x, y) = (xy^2, x^2y)$ in $g(x, y)$:

$$Dg(f(x, y)) = \begin{pmatrix} e^{xy^2} \sin(x^2y) & e^{xy^2} \cos(x^2y) \end{pmatrix}$$

Nu de kettingregel toepassen:

$$D(g \circ f)(x, y) = \begin{pmatrix} e^{xy^2} (y^2 \sin(x^2y) + 2xy \cos(x^2y)) & e^{xy^2} (2xy \sin(x^2y) + x^2 \cos(x^2y)) \end{pmatrix}$$

References